the nucleus. The kinetic energy in the interior of the nucleus was obviously still greater; its value at any given point is indicated by the separation of the horizontal straight line from the potential energy curve. The illustration suggests that the particle vibrates to and fro in the interior and bounces backwards and forwards, one might say, from one side of the ‘potential container’ to the other. The energy in the interior, shown by the straight line, is always the sum total of its kinetic and potential energies.

At first glance it seems to be impossible to see clearly how this particle can emerge from the nucleus at all. For according to the concepts of ordinary mechanics, it would not be able to move any further outward than the sides of the ‘potential container’, for at the point where the straight line crosses the potential curve, the kinetic energy vanishes—the particle comes to rest. The energy at its disposal does not enable it to traverse the potential barrier which separates the interior from the outside space. Thus according to classical mechanics, a decay reaction could not take place at all. The potential barrier would guarantee the stability of the nucleus. It could, of course, be assumed that the nucleus contains other particles as well which are likewise in motion, and that these particles might transfer energy to the alpha particle to help the latter cross the potential barrier. But this can occur only when the nucleus is in an excited state—when it has a surplus of energy. In the normal state of the atom, no such free energy is available. For the energy which the particles still possess is—according to the uncertainty principle—zero-point energy, and this energy cannot be utilized, nor can it be transferred to other particles.

This is where wave mechanics comes to the rescue. The movement of alpha particles is governed by the laws of wave mechanics and quantum mechanics, not by the laws of ordinary mechanics. On the ground of the repeatedly mentioned wave-particle duality, we can think, instead of a particle vibrating back and forth within the nucleus, of a wave which is reflected back and forth from the walls of the ‘potential container’, and we can even imagine it as a stationary wave. But now we must explain how such a wave can be reflected from these walls at all. Obviously, this phenomenon is based on a process for which we can find an analogy in the total reflection of light at the surface separating two transparent refractive substances, such as, for instance, the boundary plane between glass and air. This phenomenon occurs, for instance, in the prisms of a Zeiss field glass. If light falls perpendicularly on one of the short sides of a right-angled triangular glass prism, it will strike the surface of its hypotenuse at an angle of 45 deg. However, according to the law of refraction, it cannot emerge from this surface by refraction, but will be totally reflected in the same way as it would be by a perfectly reflecting mirror. (Figure 22 (*a*).)